

Collector Cabinet Onboarding Requirements

Project: Luciteria Collector Cabinet
Document Version: 1.0
Last Updated: May 26, 2026
Status: Requirements Specification

1. Executive Summary

1.1 Purpose

The Collector Cabinet onboarding experience is designed to seamlessly introduce two distinct user types to a unified collection management platform:

- 1. **Cube of the Month Subscribers** – Customers with active element subscriptions who need to track ownership, manage formats, and ensure intelligent assignment of future shipments
- 2. **Independent Collectors** – Element enthusiasts who want to catalog their existing collections, track missing elements, and share wishlists with gift-givers

The onboarding flow must accomplish three critical objectives:

- **Identity Verification** – Authenticate users via Shopify customer accounts
- **Collection Initialization** – Gather existing ownership data to populate the collector dashboard
- **Subscription Detection** – Identify active subscribers and link them to subscription management features

1.2 Target Audiences

Audience	Primary Goal	Key Pain Point	Success Metric
New Subscribers	Prevent duplicate shipments	Manual communication with support	Zero duplicate shipments after onboarding
Existing Subscribers	Retroactively prevent duplicates	Already own elements in subscription queue	Collection data captured within 48 hours of launch
Independent Collectors	Organize and share collections	Spreadsheet chaos or no tracking	Wishlist shared with 1+ people
Gift Buyers	See recipient’s wishlist	Guessing what they already own	View-only access to wishlist (Phase 2)

1.3 Core Onboarding Flow

Entry Point → Shopify Auth → Welcome Screen → Collection Format Selection → Manual Collection Entry → Dashboard Preview → Subscription Detection (async) → Complete (Dashboard Access)

Total Time Estimate: 3-8 minutes depending on collection size

Critical Path: Auth → Format Selection → Dashboard Access

Optional Path: Collection Entry (can be skipped and completed later)

2. Prerequisites

2.1 Shopify Configuration Required Before Launch

A. Subscription Products Setup

Product Structure:

Product Type: Subscription
Billing Frequency: Monthly

Product Name	SKU Pattern	Description	Price
Cube of the Month (10mm)	SUB-10MM	Monthly 10mm element cube	TBD
Cube of the Month (25.4mm)	SUB-25MM	Monthly 25.4mm element cube	TBD
Cube of the Month (50mm)	SUB-50MM	Monthly 50mm element cube	TBD
Cube of the Month (Lucite Cubes)	SUB-LUCITE	Monthly Lucite cube element	TBD
Cube of the Month (Ampoules)	SUB-AMPOULE	Monthly ampoule element	TBD

Required Metadata Fields:

- `collection_format` - Maps to CollectionFormat enum (TEN_MM, TWENTY_FIVE_MM, etc.)
- `subscription_type` - "cube_of_month"
- `auto_assignment_enabled` - true/false

Implementation Note: Product IDs must be added to `/app/config/shopify.js` after creation.

B. Customer Metafields Schema

```
// Namespace: collector_cabinet
{
  "has_onboarded": "boolean",
  "onboarding_completed_at": "date_time",
  "collection_format_preference": "single_line_text_field",
  "wishlist_share_token": "single_line_text_field"
}
```

C. App Installation & Permissions

Required Shopify Scopes:

- `read_customers` - Authenticate users and read customer data
- `write_customers` - Store onboarding state in customer metafields
- `read_orders` - Verify past element purchases for collection pre-fill
- `read_products` - Access SKU mapping and subscription product details
- `read_subscriptions` / `write_subscriptions` - Detect and manage active subscriptions

D. Email Template Configuration

Template ID Required:

- **Wishlist Share Email** - Sent when user generates shareable wishlist link
- **Subscription Welcome Email** - Include Collector Cabinet CTA
- **Post-Purchase Email** - Include “Add to Collection” CTA

3. User Journeys

3.1 Journey A: New Cube of the Month Subscriber

Persona: Sarah, 34, science teacher

Context: Just purchased 25.4mm subscription, receives welcome email

Timeline

1. Day 0 (Purchase Day)

- Completes checkout for subscription
- Receives order confirmation + separate welcome email
- Welcome email contains: “Start Your Collector Cabinet” CTA button

2. Day 0+30min (First App Access)

- Clicks CTA → Redirects to app with Shopify SSO
- **Screen 1:** Welcome splash with video preview of dashboard
- **Screen 2:** “Select Your Subscription Format” → Pre-selected to 25.4mm (detected from order)
- **Screen 3:** “Do you already own any elements?” → Options: [Start Fresh] [I Have Some Elements]
- Selects “Start Fresh” → Skips manual entry
- **Screen 4:** Dashboard preview with animated walkthrough
- Lands on empty dashboard with tooltips

3. Background Process:

- System detects subscription via Shopify API (async, ~5min delay)
- Updates user record with `subscription_status: ACTIVE`

- Triggers assignment engine to queue first shipment
- Sends confirmation: "Your first shipment will be [Element Name] on [Date]"

User Story

AS A new subscriber
 I WANT TO see my upcoming shipment immediately after onboarding
 SO THAT I know the system recognizes my subscription
 AND I trust it won't send duplicates

3.2 Journey B: Existing Collector (No Subscription)

Persona: Michael, 52, engineer

Context: Owns 30+ elements purchased over 5 years, finds app via product page banner

Timeline

1. Discovery

- Browsing product page for Osmium cube
- Sees banner: "Track Your Collection – Never Buy Duplicates"
- Clicks → App opens with Shopify SSO

2. Onboarding Flow

- **Screen 1:** Welcome – "Join 1,000+ Element Collectors"
- **Screen 2:** "What format do you collect?" → Selects 25.4mm
- **Screen 3:** "Let's add your existing collection"
 - Periodic table appears with all elements clickable
 - Clicks owned elements → They turn green with checkmark
 - Progress indicator: "12/118 elements marked"
 - **Screen 4:** "Want to create a wishlist?" → Explains sharing feature
 - Clicks 10 missing elements → They turn yellow with star icon
 - Generates shareable link: `luciteria.com/wishlist/a8f9e2c1`

3. Post-Onboarding

- Dashboard shows:
 - 12 owned (green)
 - 10 wishlisted (yellow)
 - 96 missing (gray)
 - Shares wishlist link with spouse via email
 - Returns monthly to update collection as he buys more

User Story

AS AN independent collector
 I WANT TO mark my existing elements quickly
 SO THAT I can see what I'm missing at a glance
 AND share my wishlist with gift-givers

3.3 Journey C: Subscriber with Existing Collection

Persona: Elena, 28, chemistry PhD student

Context: Already owns 45 elements (mixed formats), just subscribed to 10mm format

Timeline

1. Onboarding Entry

- Post-purchase email CTA → App opens
- **Screen 2:** Format selection → Selects 10mm (matches subscription)
- Warning appears: “We noticed you’ve purchased elements before. Import your order history?”

2. Collection Pre-Fill (Optional)

- System scans past orders for element SKUs
- Pre-populates 18 elements in 10mm format
- **Screen 3:** “We found these in your order history” → Review table
 - User reviews and confirms 16, removes 2 (were gifts)

3. Manual Addition

- **Screen 4:** “Add any elements from other sources”
- Clicks 27 additional elements (bought from other vendors)
- Total: 43 elements marked

4. Assignment Preview

- System shows: “Based on your collection, your first shipment will be: **Scandium**”
- Tooltips explain: “We skip elements you already own”

Edge Case Handling

- **Multi-format ownership:** User owns 50mm elements but subscribing to 10mm
- System filters order history to only show 10mm SKUs
- Prompts: “Want to track other formats too? (Coming in Phase 2)”

4. Onboarding Screens (Detailed Specifications)

Screen 0: Entry Point Authentication

Trigger: User clicks any CTA leading to Collector Cabinet

Technical Flow:

1. App receives request at /auth/login
2. Redirect to Shopify OAuth endpoint
3. Shopify authenticates customer
4. Callback to /auth/callback with customer token
5. App creates/loads user record
6. Check: `has_onboarded === true?`
 - YES ☐ Redirect to /dashboard
 - NO ☐ Continue to Screen 1

Error Handling:

- **Not Logged In:** Redirect to Shopify customer login page
- **Auth Failure:** Show error + retry button
- **No Customer Account:** Prompt to create Shopify account first

Screen 1: Welcome Splash

Visual Design:

- Full-screen hero image: Periodic table with glowing owned elements
- Centered card overlay (max-width: 600px)

Copy:

Welcome to Your Collector Cabinet

Track your element collection, prevent duplicates,
and share your wishlist with friends and family.

[Watch 30s Demo Video] [Get Started →]

Video Content (Optional):

- 0:00-0:10 – Dashboard pan showing colorful periodic table
- 0:10-0:20 – Closeup of clicking elements to mark ownership
- 0:20-0:30 – Sharing wishlist link via email

Implementation Notes:

- Video hosted on Shopify CDN or YouTube (unlisted)
 - “Skip” option in top-right corner
 - Progress indicator: “Step 1 of 4”
-

Screen 2: Collection Format Selection

Purpose: Determine which SKU universe to filter (critical for subscribers)

Layout:

- Centered heading + 5 large cards (responsive grid)

Copy:

What format **do** you collect?

Each subscription focuses on one size. Choose the format
you're **collecting**, and we'll **show** you relevant products.

[Card: 10mm Cubes]

[Image: Tiny cube]

[Card: 25.4mm Cubes]

[Image: Standard cube] ← Selected by default **if**

subscription detected

[Card: 50mm Cubes]

[Image: Large cube]

[Card: Lucite Cubes]

[Image: Acrylic block]

[Card: Ampoules]

[Image: Glass vial]

[**Continue** →]

Interaction Logic:

1. Subscription Pre-Detection (If Available):

- If user has active subscription, pre-select matching format

- Show banner: “We’ve selected 25.4mm based on your subscription”
- Allow user to change if incorrect

1. **Multi-Format Collections (Phase 2):**

- Show link: “Collect multiple formats? Contact us for beta access”

2. **Validation:**

- Cannot proceed without selecting exactly 1 format
- Selected card highlights with border + checkmark

Database Update:

- Save `collection_format` to User record
- Log event: `onboarding_format_selected`

Screen 3: Existing Collection Entry

Purpose: Populate initial ownership data

Layout:

- Interactive periodic table (full-screen)
- Right sidebar with instructions + progress

Copy (Sidebar):

Add Your Existing Collection

Click any element you already own in [10mm] format.
Don't worry—you can always update this later.

Progress: 0/118 elements marked

[Mark All as Owned] ← Rare but useful

[Skip This Step]

[Continue →] ← Enabled after any selection OR skip

Periodic Table Interaction:

- **Idle State:** Gray tiles with element symbol + name
- **Hover State:** Tooltip shows element name + atomic number
- **Clicked (Owned):** Tile turns green, checkmark icon appears
- **Clicked Again (Undo):** Returns to gray

Smart Features:

1. **Bulk Selection (Optional):**

- “Select by group” dropdown: Alkali Metals, Noble Gases, etc.
- Click group → All elements in category toggle to owned

1. **Search Bar:**

- Type “Gold” → Au tile highlights
- Click → Marks as owned

2. **Order History Import (If Available):**

- Button: “Import from Past Orders”
- Modal shows table of detected purchases:

Element	Format	Order Date	Action
Iron	10mm	2024-03-12	[✓ Add]
Copper	10mm	2024-05-20	[✓ Add]

- User can bulk-add or review individually

Validation:

- No minimum required (skip is allowed)
- Maximum: 118 elements (can't mark more than exist)

Database Update:

- Create `OwnedElement` records for each marked element
- Fields: `user_id`, `atomic_number`, `collection_format`, `acquisition_source: MANUAL_ENTRY`, `created_at`

Screen 4: Wishlist Setup (Optional)

Purpose: Introduce wishlist feature and capture desired elements

Trigger: Only shown if user marked < 50 elements (assumes they have gaps to fill)

Copy:

Create Your Wishlist

Mark elements you'd love to receive as gifts.
We'll generate a shareable link for friends and family.

[Same Periodic Table Interface]

Now showing:

- Green (✓) = Already Own
- Yellow (★) = Wishlist
- Gray = Not Tracked

[Skip Wishlist]

[Generate Shareable Link ➔]

Interaction Logic:

- Clicking owned (green) element does nothing
- Clicking gray element marks as wishlist (yellow)
- Clicking yellow element removes from wishlist (back to gray)
- "Generate Link" button appears after 1+ wishlist item

Shareable Link Generation:

1. User clicks "Generate Shareable Link"
2. System creates unique token: `UUID.v4()`
3. Stores in database:

```
javascript
{
  user_id: 123,
  share_token: "a8f9e2c1-b8d3-4f5e-9a1c-d2e3f4a5b6c7",
  expires_at: null, // Never expires in V1
}
```



```
view_count: 0
```

```
}
```

4. Displays modal with:

```
...
```

Your Wishlist is Ready to Share!

Copy this link and send it to anyone:

<https://luciteria.com/wishlist/a8f9e2c1>

[Copy Link] [Send via Email]

They'll see your wishlist and can shop directly from it.

```
...
```

Email Share Option:

- Opens email client with pre-filled message:

```
...
```

Subject: My Element Collection Wishlist

Hi!

I'm building a collection of the periodic table. Here's my wishlist of elements I'd love to receive:

<https://luciteria.com/wishlist/a8f9e2c1>

You can view what I already own and what I'm still looking for.

Thanks!

```
...
```

Implementation Notes:

- Wishlist is view-only in V1 (no gift purchasing flow)
- Recipient sees public periodic table with legend
- Track `view_count` for analytics

Screen 5: Completion & Dashboard Preview

Purpose: Celebrate completion and orient user to dashboard

Layout:

- Success animation (confetti or element cascade)
- Centered card with summary stats

Copy:

Your Collector Cabinet is Ready! 🎉

Collection Summary:

- [12] Elements Owned
- [8] Elements Wishlisted
- [98] Elements to Discover

[View Your Dashboard →]

If Subscriber Detected (Async):

- Additional panel appears:

```
...
```

 Your Next Shipment

Element: Scandium (Sc)

Ships: June 15, 2026

Format: 25.4mm Cube

We automatically skip elements you own.

[View Shipment Calendar]

```
...
```

First-Time User Tooltips (On Dashboard):

- Pointer to periodic table: "Click any element to update ownership"
- Pointer to filters: "Switch between Owned, Missing, or Wishlisted"
- Pointer to profile: "Manage your subscription here"

Database Update:

- Set `has_onboarded: true`
- Set `onboarding_completed_at: NOW()`
- Log event: `onboarding_completed`

5. Technical Requirements

5.1 Authentication & Session Management

Flow:

```
// /app/routes/auth/login.js
export async function loader({ request }) {
  const shopDomain = getShopDomain(request);
  const redirectUri = `${APP_URL}/auth/callback`;

  return redirect(
    `https://${shopDomain}/admin/oauth/authorize?` +
    `client_id=${SHOPIFY_API_KEY}&` +
    `scope=${REQUIRED_SCOPES.join(',')}&` +
    `redirect_uri=${redirectUri}`
  );
}
```

Session Storage:

- Store encrypted session token in secure HTTP-only cookie
- Session duration: 30 days (refresh on activity)
- Customer ID stored in session for database queries

Security:

- CSRF protection via Shopify nonce validation
 - No plaintext tokens in client-side storage
 - Rate limiting: 10 auth attempts per IP per hour
-

5.2 Subscription Detection

Challenge: Subscription data may not be immediately available after purchase

Solution: Async Detection with Polling

```
// /app/lib/subscription-detector.js
export async function detectSubscription(userId) {
  const customer = await getShopifyCustomer(userId);

  // Check for active subscription contracts
  const subscriptions = await shopify.subscriptionContracts.list({
    customer_id: customer.shopify_id
  });

  const activeSubscription = subscriptions.find(sub =>
    sub.status === 'ACTIVE' &&
    SUBSCRIPTION_PRODUCT_IDS.includes(sub.product_id)
  );

  if (activeSubscription) {
    // Extract format from product metadata
    const format = parseSubscriptionFormat(activeSubscription.product_id);

    await db.user.update({
      where: { id: userId },
      data: {
        subscription_status: 'ACTIVE',
        subscription_format: format,
        subscription_detected_at: new Date()
      }
    });

    // Trigger first assignment
    await assignmentEngine.queueNextShipment(userId);

    return true;
  }

  return false;
}
```

Polling Strategy:

- **Immediate:** Check on onboarding completion
- **Retry 1:** 5 minutes after onboarding
- **Retry 2:** 30 minutes after onboarding
- **Retry 3:** 24 hours after onboarding
- **Manual Trigger:** “Refresh Subscription Status” button in profile

User Communication:

- If not detected after 24 hours:
- ...

We’re still confirming your subscription.
This usually takes a few minutes.

[Refresh Status] [Contact Support]
...

5.3 Data Storage Schema

Core Tables:

```

model User {
  id String @id @default(uuid())
  shopifyCustomerId String @unique
  email String
  firstName String?
  lastName String?

  // Onboarding state
  hasOnboarded Boolean @default(false)
  onboardingCompletedAt DateTime?
  onboardingStep Int @default(0) // Track partial progress

  // Collection preferences
  collectionFormat CollectionFormat?

  // Subscription data
  subscriptionStatus SubscriptionStatus @default(NONE)
  subscriptionFormat CollectionFormat?
  subscriptionDetectedAt DateTime?
  shopifySubscriptionId String?

  // Relationships
  ownedElements OwnedElement[]
  wishlistItems WishlistItem[]
  wishlistShare WishlistShare?
  shipments Shipment[]

  createdAt DateTime @default(now())
  updatedAt DateTime @updatedAt
}

model OwnedElement {
  id String @id @default(uuid())
  userId String
  user User @relation(fields: [userId], references: [id])

  atomicNumber Int
  collectionFormat CollectionFormat
  acquisitionSource AcquisitionSource @default(MANUAL_ENTRY)
  acquiredAt DateTime @default(now())

  // Optional: Link to original order
  shopifyOrderId String?
  shopifyLineItemId String?

  @@unique([userId, atomicNumber, collectionFormat])
}

model WishlistItem {
  id String @id @default(uuid())
  userId String
  user User @relation(fields: [userId], references: [id])
  atomicNumber Int
  priority Int @default(0) // For ranking (not used in V1)
  addedAt DateTime @default(now())

  @@unique([userId, atomicNumber])
}

model WishlistShare {
  id String @id @default(uuid())
  userId String @unique

```

```

user      User      @relation(fields: [userId], references: [id])
shareToken String   @unique @default(uuid())
viewCount Int       @default(0)
lastViewedAt DateTime?
createdAt DateTime @default(now())

model OnboardingEvent {
  id          String @id @default(uuid())
  userId      String
  eventType   String // "format_selected", "collection_entry_started", etc.
  metadata    Json?
  createdAt  DateTime @default(now())

  @@index([userId, createdAt])

enum AcquisitionSource {
  MANUAL_ENTRY
  SUBSCRIPTION_SHIPMENT
  ORDER_IMPORT
  PURCHASED

enum SubscriptionStatus {
  NONE
  ACTIVE
  PAUSED
  CANCELLED

enum CollectionFormat {
  TEN_MM
  TWENTY_FIVE_MM
  FIFTY_MM
  LUCITE_CUBES
  AMPOULES

```

5.4 Performance Requirements

Metric	Target	Measurement
Auth Redirect	< 2 seconds	Time from CTA click to welcome screen
Periodic Table Render	< 500ms	Initial paint of 118 elements
Element Toggle	< 50ms	Click to visual feedback
Onboarding Completion	< 1 second	Final screen load + DB write
Subscription Detection	< 5 minutes	95% detected within 5min of purchase

Optimization Strategies:

- Periodic table: SVG sprite sheet (single HTTP request)
- Database: Batch insert owned elements (not 1 query per element)
- CDN: Static assets served from Shopify CDN
- Caching: User session data cached in Redis (if available)

6. Entry Points

6.1 Primary Entry Points

A. Post-Purchase Email (Subscribers Only)

Trigger: Order confirmed with subscription product

Email Template:

Subject: Welcome to Cube of the Month! 

Hi [First Name],

Thanks **for** subscribing! Your first element ships on [Date].

To make sure we never send you duplicates, set up your Collector Cabinet:

[Start Your Collector Cabinet 

This takes 2 minutes and helps us customize your shipments.

Questions? Reply to this email.

The Luciteria Team

CTA Link: https://luciteria.com/apps/collector-cabinet?source=welcome_email

Implementation:

- Shopify Flow automation or custom webhook
- Trigger: `orders/create` with subscription line item
- Template stored in Shopify Email or external ESP

B. Shopify Customer Account Menu

Location: Customer account dashboard navigation

Menu Item:

```

My Account
├─ Order History
├─ Subscriptions
├─ Collector Cabinet ← NEW
└─ Account Details

```

Implementation:

```

<!-- theme.liquid -->
<nav class="customer-account-menu">
  <a href="/apps/collector-cabinet">
    <svg><!-- Periodic table icon --></svg>
    Collector Cabinet
    {% if customer.metafields.collector_cabinet.has_onboarded == false %}
      <span class="badge">New</span>
    {% endif %}
  </a>
</nav>

```

C. Product Page Banner (Independent Collectors)

Placement: Above “Add to Cart” button on all element product pages

Banner Copy:

[Icon] Already collecting? Track your collection **to** avoid duplicates.
[Set Up Collector Cabinet →]

Conditional Display:

- Show only if customer is NOT a subscriber
- Hide after user completes onboarding
- A/B test: Show vs. don't show impact on conversions

Implementation:

```

<!-- product-template.liquid -->
{% unless customer.metafields.collector_cabinet.has_onboarded %}
  <div class="collector-cabinet-banner">
    <p>Already collecting? Track your collection to avoid duplicates.</p>
    <a href="/apps/collector-cabinet?source=product_page">Set Up Cabinet</a>
  </div>
{% endunless %}

```

D. Website Navigation (Global)

Location: Main site header (desktop) or hamburger menu (mobile)

Placement:

[Logo] Shop | Subscriptions | Collector Cabinet | About [Cart] [Account]
^^

Visibility:

- Show for all logged-in customers
- Show “Login to Access” for guest visitors

E. Subscription Confirmation Page

Placement: Thank-you page after completing subscription checkout

Hero Section:

 You're Subscribed!

Order #12345 confirmed. Your first cube ships June 15.

Next Step: Set up your Collector Cabinet to prevent duplicates.

[\[Set Up Now →\]](#) [\[I'll Do This Later\]](#)

Technical Implementation:

```
<!-- checkout-thank-you.liquid -->
{% if order.line_items contains 'subscription' %}
  <div class="collector-cabinet-cta">
    <h2>Set Up Your Collector Cabinet</h2>
    <p>Tell us what you already own—takes 2 minutes.</p>
    <a href="/apps/collector-cabinet?source=checkout_thankyou&order={{ order.id }}">
      Set Up Now
    </a>
  </div>
{% endif %}
```

6.2 Entry Point Tracking

URL Parameters:

All entry point links include `?source=` parameter for analytics

Source Value	Description
welcome_email	Post-purchase subscriber email
account_menu	Customer account dashboard
product_page	Banner on element product pages
main_nav	Website header/navigation
checkout_thankyou	Post-subscription confirmation page
support_link	Customer support directed user

Database Logging:

```
// On app entry
await db.onboardingEvent.create({
  data: {
    userId: user.id,
    eventType: 'onboarding_entry',
    metadata: {
      source: request.query.source,
      referrer: request.headers.referrer,
      timestamp: new Date()
    }
  }
});
```

Analytics Dashboard (Admin):

- “Top Entry Points” chart
- Conversion rate by source
- Time-to-onboard by source

7. Feature Access Matrix

7.1 Core Principle: No Feature Paywalls

Philosophy: All users (subscribers and independent collectors) have access to the same core features. The distinction is in **automated workflows**, not **manual capabilities**.

Feature	Subscribers	Independent Col-lectors	Notes
View Dashboard	✓ Full Access	✓ Full Access	Both see 118-element periodic table
Mark Owned Elements	✓ Unlimited	✓ Unlimited	No caps or restrictions
Create Wishlist	✓ Unlimited	✓ Unlimited	Same sharing capabilities
Share Wishlist Link	✓ Generate Link	✓ Generate Link	Public view for recipients
Track Multiple Formats	⚠ Phase 2	⚠ Phase 2	V1: One format per user
Auto-Assignment	✓ EXCLUSIVE	✗ N/A	System queues monthly shipments
Shipment Calendar	✓ View Upcoming	✗ N/A	See 12 months of assignments
Skip/Swap Shipments	⚠ Phase 2	✗ N/A	Manual override of assignments
Subscription Management	✓ Pause/Cancel	✗ N/A	Links to Shopify subscription portal
Order History Import	✓ Available	✓ Available	Both can scan past orders
Manual Collection Entry	✓ Via Dashboard	✓ Via Dashboard	Identical UI
CSV Export	✓ Download Data	✓ Download Data	Export owned/wishlist

7.2 Subscriber-Exclusive Features (Automated Workflows)

A. Auto-Assignment Engine

What It Does:

- Scans owned elements to prevent duplicates
- Respects wishlist priority (Phase 2)
- Excludes out-of-stock items (sequence shifting)
- Excludes precious metals (manual approval required)
- Generates monthly shipment queue

User Experience:**[Dashboard Panel]**

 Your Next Shipments

June 2026: Scandium (Sc) - Ships June 15

July 2026: Yttrium (Y) - Ships July 15

August 2026: Pending Assignment

Access Control:

```
// Only show if user.subscriptionStatus === 'ACTIVE'
{user.subscriptionStatus === 'ACTIVE' && (
  <ShipmentCalendar userId={user.id} />
)}
```

B. Duplicate Prevention Alerts**What It Does:**

- If user manually marks an element as owned that was already assigned
- Show warning: “Scandium was scheduled for your next shipment. We’ll skip it.”
- Auto-reassign next element

Implementation:

```
async function markElementOwned(userId, atomicNumber) {
  // Check if element is in upcoming shipment queue
  const upcomingShipment = await db.shipment.findFirst({
    where: {
      userId,
      atomicNumber,
      status: 'ASSIGNED',
      shipDate: { gte: new Date() }
    }
  });

  if (upcomingShipment) {
    // Trigger reassignment
    await assignmentEngine.reassignShipment(upcomingShipment.id);

    // Show user notification
    return {
      success: true,
      message: `${elementName} was in your queue. We've updated your shipments.`
    };
  }
}
```

7.3 Independent Collector Experience

Key Point: Non-subscribers see the exact same dashboard UI, but without subscription-related panels.

What They See:

- Periodic table with owned/wishlist/missing states
- Collection statistics (12 owned, 8 wishlisted, 98 missing)
- Wishlist sharing link
- “Upgrade to Subscription” CTA (soft sell)

What They DON’T See:

- Shipment calendar
- “Next shipment” panel
- Subscription management settings

Upgrade CTA Placement:

[Dashboard - Bottom of Sidebar]

 Want monthly surprises?

Subscribe to Cube of the Month and get a new element every month—automatically skipping ones you own.

[Explore Subscriptions →]

Conversion Tracking:

- Track clicks on “Explore Subscriptions” CTA
- Attribute subscription purchases to Collector Cabinet if within 7 days

8. Wishlist Implementation

8.1 Feature Overview

Purpose: Enable collectors to share desired elements with friends and family for gift-giving

Key Clarification: Wishlist does NOT influence subscription shipment selection in V1. It is purely for sharing with external gift-givers.

8.2 User Flow: Creating a Wishlist

Step 1: Mark Wishlist Items

From dashboard:

- User clicks any gray (missing) element
- Dropdown appears:
 - [+ Mark as Owned]
 - [★ Add to Wishlist]
- Selecting “Add to Wishlist” turns element yellow with star icon

Step 2: Generate Share Link

- User clicks “Share Wishlist” button in sidebar

- Modal appears:

...

Share Your Wishlist

[Generate Shareable Link]

Anyone with this link can view your wishlist.

They'll see which elements you already own (so they don't buy duplicates) and which you're hoping for.

- Clicking "Generate" creates unique URL:

<https://luciteria.com/wishlist/a8f9e2c1>

...

Step 3: Share Options

Your Wishlist Link is Ready!

<https://luciteria.com/wishlist/a8f9e2c1>

[Copy Link] [Send via Email] [Download PDF]

- **Copy Link:** Copies to clipboard with toast notification
- **Send via Email:** Opens default email client with template
- **Download PDF:** Generates printable wishlist (Phase 2)

8.3 Public Wishlist View (Recipient Experience)

URL Structure: `/wishlist/:shareToken`

Layout:

- Public-facing page (no login required)
- Header: "[Collector Name]'s Element Wishlist"
- Periodic table with three states:
- **Green (✓):** Already owned (don't buy)
- **Yellow (★):** Wishlist items
- **Gray:** Not tracked

Copy:

Michael's Element Collection

Owned: 12 elements

Wishlist: 8 elements

Click any yellow element to shop for it on Luciteria.com

Interaction:

- Clicking a yellow (wishlist) element → Redirects to product page
- URL: `/products/scandium-10mm-cube?ref=wishlist_a8f9e2c1`
- Tracks conversions via `ref` parameter

Privacy Considerations:

- No personal information shown (just first name)
 - No email address or account details
 - No subscription status visible
 - Cannot edit collection (view-only)
-

8.4 Wishlist Synchronization

Challenge: Recipient buys element as gift → Collector receives it → Needs to mark as owned → Wishlist should update for other potential gift-givers

Solution: Real-Time Updates

```
// When collector marks element as owned
async function markElementOwned(userId, atomicNumber) {
  await db.ownedElement.create({ userId, atomicNumber });

  // Remove from wishlist if present
  await db.wishlistItem.deleteMany({
    where: { userId, atomicNumber }
  });

  // Trigger WebSocket broadcast (if using real-time)
  websocket.broadcast(`wishlist:${user.wishlistShare.shareToken}`, {
    action: 'element_acquired',
    atomicNumber
  });
}
```

User Experience:

- Recipient viewing wishlist sees element disappear from yellow list
 - Toast notification: “This element was just added to their collection”
-

8.5 Edge Cases

Case 1: Wishlist Link Shared Before Generation

Scenario: User marks wishlist items but never clicks “Generate Link”

Behavior:

- Wishlist exists in database but no `shareToken` created
 - “Share Wishlist” button visible in dashboard
 - Clicking generates token retroactively
-

Case 2: User Deletes All Wishlist Items

Scenario: User removes all yellow stars → Wishlist is empty

Behavior:

- Share link remains valid but shows:

Michael's wishlist is currently empty.

Check back later!

Case 3: User Marks Entire Periodic Table as Owned

Scenario: User owns all 118 elements

Behavior:

- Wishlist automatically disabled
- Dashboard shows: "Collection Complete! 🎉"
- Share link shows:
...

Michael has completed the entire periodic table!
Consider a gift card instead.

[Shop Luciteria Gift Cards →]
...

Case 4: Non-Customer Views Wishlist

Scenario: Recipient doesn't have a Luciteria account

Behavior:

- Can view wishlist without login
- Clicking element → Redirected to product page
- Product page shows normal guest checkout flow
- Attribution tracked via `?ref=wishlist_token`

9. Edge Cases & Error Handling

9.1 Subscription-Related Edge Cases

Edge Case 1: Subscription Cancelled Mid-Onboarding

Scenario:

1. User purchases subscription
2. Starts onboarding
3. Cancels subscription before completing onboarding
4. Completes onboarding later

Expected Behavior:

- User can complete onboarding (keep collection data)
- No shipment calendar shown
- Dashboard shows:
...

Your subscription was cancelled.
Your collection data is safe—you can still track manually.

[Resubscribe] [Continue as Independent Collector]
...

Implementation:

```
// Check subscription status on every dashboard load
const activeSubscription = await getActiveSubscription(userId);
if (!activeSubscription && user.subscriptionStatus === 'ACTIVE') {
  await db.user.update({
    where: { id: userId },
    data: { subscriptionStatus: 'CANCELLED' }
  });
}
```

Edge Case 2: User Subscribes After Completing Onboarding**Scenario:**

1. User onboards as independent collector
2. Weeks later, subscribes to Cube of the Month

Expected Behavior:

- Subscription detected via polling (async)
- Dashboard auto-updates to show shipment calendar
- First shipment assigned based on existing owned elements
- Welcome message:

...

Welcome to Cube of the Month!

We've analyzed your collection (12 elements owned) and assigned your first shipment: Scandium (June 15).

[View Shipment Calendar →]

...

Implementation:

- Webhook listener for `subscription_contracts/create`
- Triggers subscription detection + assignment engine
- Sends in-app notification

Edge Case 3: User Changes Subscription Format**Scenario:**

1. User subscribes to 10mm format
2. Onboards and marks 20 owned 10mm elements
3. Cancels 10mm subscription and subscribes to 25.4mm

Expected Behavior (V1 Limitation):

- User must re-onboard for new format
- Dashboard shows warning:

...

You've switched to a new format (25.4mm).

Please update your collection for this format.

[Start Collection Entry for 25.4mm →]

...

- Owned 10mm elements remain in database but hidden
- Fresh start for 25.4mm format

Phase 2 Solution:

- Multi-format tracking
- User can toggle between formats in dashboard

9.2 Authentication Edge Cases

Edge Case 4: User Not Logged Into Shopify

Scenario: User clicks CTA but no active Shopify session

Expected Behavior:

1. Redirect to Shopify customer login page
2. After login, redirect back to Collector Cabinet with onboarding flow intact
3. Preserve `?source=` parameter for analytics

Implementation:

```
// Store intended destination in session
export async function loader({ request }) {
  const session = await getSession(request);
  if (!session.customerId) {
    // Store return URL
    session.set('returnTo', request.url);

    return redirect('/auth/login');
  }
}
```

Edge Case 5: User Has No Shopify Customer Account

Scenario: First-time visitor clicks Collector Cabinet CTA

Expected Behavior:

- Redirect to Shopify account creation page
- After account creation, redirect to onboarding
- Show welcome message:

...

Welcome to Luciteria!

You've created your account. Now let's set up your Collector Cabinet.

...

9.3 Data Integrity Edge Cases

Edge Case 6: User Marks Element Twice (Duplicate Entry)

Scenario: Bug or race condition causes same element marked twice

Prevention:

- Database unique constraint: `@@unique([userId, atomicNumber, collectionFormat])`

- Graceful error handling:

```
javascript
try {
  await db.ownedElement.create({ userId, atomicNumber });
} catch (error) {
  if (error.code === 'P2002') { // Prisma unique constraint error
    return { success: true, message: 'Already in your collection' };
  }
  throw error;
}
```

Edge Case 7: User Tries to Wishlist Owned Element

Scenario: User accidentally clicks “Add to Wishlist” on green (owned) element

Prevention:

- UI: Disable wishlist option for owned elements

- Backend validation:

```
````javascript
const alreadyOwned = await db.ownedElement.findUnique({
 where: { userId_atomicNumber: { userId, atomicNumber } }
});

if (alreadyOwned) {
 return { error: 'You already own this element' };
}
````
```

Edge Case 8: User Marks All 118 Elements on First Login

Scenario: Suspicious activity or bot

Detection + Handling:

- Track onboarding duration

- If < 30 seconds and 100+ elements marked:

...

Whoa, that was fast!

Can you confirm you own all these elements?

This helps us prevent errors.

[Yes, I Own Them All] [Let Me Review]

...

- Log event for admin review

9.4 Wishlist Edge Cases

Edge Case 9: Recipient Buys Element, But Collector Doesn't Update

Scenario:

1. Grandma views wishlist, buys Scandium
2. Collector receives Scandium but doesn't mark as owned
3. Grandpa views wishlist, also buys Scandium

Mitigation (Phase 2):

- Email collector after gift purchase:

```
```
```

Did you receive [Element Name]?

[Yes, Mark as Owned] [Not Yet]

```
```
```

- Track purchase attribution via `?ref=wishlist_token``
- Temporarily gray out element on wishlist for 7 days

V1 Behavior:

- No automatic updates
- Relies on collector manually updating

Edge Case 10: User Generates Multiple Share Links

Scenario: User clicks "Generate Link" multiple times

Expected Behavior:

- First click: Creates `shareToken`
- Subsequent clicks: Returns same token (doesn't create new)
- Implementation:

```
```javascript
```

```
async function getOrCreateShareToken(userId) {
 let share = await db.wishlistShare.findUnique({ where: { userId } });
```

```
 if (!share) {
 share = await db.wishlistShare.create({
 data: { userId, shareToken: uuid() }
 });
 }

 return share.shareToken;
}
```

```
```
```

9.5 Performance Edge Cases

Edge Case 11: Slow Shopify API Response

Scenario: Subscription detection API call times out (>10 seconds)

Handling:

- Show loading state with progress message:

```
...
```

Checking for subscription...

This is taking longer than usual.

[Continue Without Waiting] [Keep Waiting]

```
...
```

- If user clicks "Continue Without Waiting":
- Complete onboarding without subscription data
- Retry detection in background (polling)

Edge Case 12: User Abandons Onboarding Mid-Flow

Scenario: User completes Screen 2, then closes browser

Recovery:

- Store `onboardingStep` in database
- Next login:

```
...
```

Welcome back!

You started setting up your Collector Cabinet.

Want to pick up where you left off?

[Continue Setup] [Start Over]

```
..
```

- If "Continue": Jump to last completed step
- If "Start Over": Reset `onboardingStep`` to 0

Implementation:

```
// Save progress after each screen
async function saveOnboardingProgress(userId, step) {
  await db.user.update({
    where: { id: userId },
    data: { onboardingStep: step }
  });
}
```

10. Success Metrics

10.1 Primary KPIs

| Metric | Target | Measurement Window | Tracking Method |
|---------------------------------------|------------|---------------------------|---|
| Onboarding Completion Rate | >80% | 30 days post-launch | % of users who reach Screen 5 |
| Time to Complete | <5 minutes | Per session | Median duration (entry → completion) |
| Collection Data Capture | >50% | Within 48 hours of signup | % of users who mark ≥1 element |
| Subscription Detection Success | >95% | Within 5 minutes | % detected without manual intervention |
| Wishlist Generation Rate | >40% | 7 days post-onboarding | % who create shareable link |
| Duplicate Shipment Prevention | 100% | Ongoing | Zero duplicates sent to onboarded users |

10.2 Onboarding Funnel Analysis

Funnel Stages:

1. Entry (CTA Click) ☐ 100% baseline
2. Auth Success ☐ Target: 95%
3. **F**ormat Selection ☐ Target: 90%
4. Collection Entry Started ☐ Target: 75%
5. **O**nboarding Completed ☐ Target: 80%

Drop-Off Analysis:

- Track where users abandon
- Heatmaps on each screen
- Exit survey for users who close app during onboarding:

...

Quick Question: Why are you leaving?

- ☐ Too long / confusing
- ☐ Don't have time now
- ☐ Not interested anymore
- ☐ Technical issue
- ☐ Other: _

...

10.3 Engagement Metrics (Post-Onboarding)

| Metric | Target | Purpose |
|-----------------------------|--------------------|--------------------------------------|
| Return Visit Rate | >60% within 7 days | Measures stickiness |
| Collection Update Frequency | >2 updates/month | Shows active usage |
| Wishlist Share CTR | >25% | Measures sharing feature adoption |
| Wishlist View Count | >3 views/link | Validates sharing value |
| Subscription Upgrade Rate | >10% | Independent collectors → subscribers |

10.4 Subscriber-Specific Metrics

| Metric | Target | Purpose |
|-----------------------------|----------------------------|------------------------------------|
| Subscription Detection Time | <5 min (median) | Measures API performance |
| First Assignment Accuracy | 100% | No duplicates in first shipment |
| Manual Override Requests | <5% | Low = assignment engine works well |
| Subscription Retention | +15% vs. non-Cabinet users | Impact on churn |

Hypothesis: Subscribers who complete onboarding have higher retention because they trust the duplicate prevention system.

10.5 Entry Point Performance

Track Conversion by Source:

| Entry Point | Expected Traffic | Expected Conversion | Tracking URL |
|---------------------|--------------------------|---------------------|-------------------------------|
| Welcome Email | High (100% of new subs) | 70% | ?
source=welcome_email |
| Account Menu | Medium | 40% | ?
source=account_menu |
| Product Page Banner | High (all product views) | 5% | ?
source=product_page |
| Main Navigation | Low | 20% | ?source=main_nav |
| Checkout Thank-You | High (100% of subs) | 60% | ?
source=checkout_thankyou |

Optimization:

- A/B test email copy for welcome email
- Test banner placement on product pages (top vs. bottom)
- Experiment with urgency (“Set up in 2 minutes” vs. “Prevent duplicates”)

10.6 Technical Performance Metrics

| Metric | Target | Alert Threshold |
|---------------------|--------------|-----------------|
| Page Load Time | <2s (p95) | >3s |
| API Response Time | <500ms (p95) | >1s |
| Error Rate | <0.5% | >2% |
| Auth Failure Rate | <2% | >5% |
| Database Query Time | <100ms (p95) | >500ms |

10.7 Business Impact Metrics

Revenue Attribution:

- Track subscription purchases within 7 days of onboarding
- Track product purchases via wishlist referrals (?ref=wishlist_token)

Customer Support Impact:

- Measure ticket volume for “I already own this element” (expect 80% reduction)
- Track time saved by support team (fewer manual collection audits)

Lifetime Value (LTV):

- Compare LTV of onboarded vs. non-onboarded subscribers
- Hypothesis: Onboarded users stay 3+ months longer

10.8 Dashboard & Reporting

Admin Analytics View:

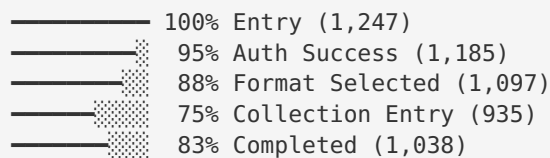
Onboarding Overview (Last 30 Days)

Total Starts: 1,247

Completed: 1,038 (83.2%) 

Avg. Time: 4m 32s

Funnel Breakdown:



Top Entry Points:

1. Welcome Email - 612 completions (59%)
2. Checkout Thank-You - 298 (29%)
3. Account Menu - 94 (9%)

Subscription Detection:

- <1 min: 72%
- <5 min: 94%
- <30 min: 98%
- >24 hours: 2% (Needs Review)

Collection Stats:

- Avg. elements marked: 8.3
- Wishlist created: 42%
- Wishlist shared: 31%

11. Implementation Phases

Phase 1: MVP (Weeks 1-4)**Scope:**

-  Shopify authentication
-  Screens 1-5 (basic onboarding)
-  Manual collection entry
-  Subscription detection (async)
-  Single-format tracking
-  Basic wishlist generation
-  Entry points: Welcome email + account menu

Launch Criteria:

- 100 beta testers complete onboarding

- <2% error rate
 - Subscription detection >90%
-

Phase 2: Enhancements (Weeks 5-8)

Scope:

- ☒ Order history import (pre-fill owned elements)
 - ☒ Multi-format tracking
 - ☒ Wishlist PDF export
 - ☒ Entry points: Product page banners
 - ☒ Advanced analytics dashboard
 - ☒ A/B testing framework
-

Phase 3: Automation (Weeks 9-12)

Scope:

- ☒ Real-time subscription detection (webhooks)
 - ☒ Wishlist purchase attribution
 - ☒ Email notifications for collection updates
 - ☒ Mobile-optimized UI
 - ☒ Bulk element selection (by group)
-

Phase 4: Advanced Features (Future)

Scope:

- Collection photography (upload images of owned elements)
 - Social sharing (public profiles)
 - Collection value tracking (resale market prices)
 - Trade/swap marketplace
 - Gamification (badges, completion rewards)
-

12. Open Questions & Decisions Needed

12.1 Pricing & Business Model

Question: Should independent collectors pay for Collector Cabinet access?

Options:

- **Free for All:** Drives adoption, converts to subscribers
- **Freemium:** Free basic tracking, \$5/month for wishlist sharing + advanced features
- **Subscriber-Only:** Only available to active subscribers (limits growth)

Recommendation: Free for all (drives subscription conversions via “upgrade” CTAs)

12.2 Data Ownership & Portability

Question: What happens to collection data if user cancels subscription?

Options:

- **Retain Forever:** Data persists, user can return anytime
- **30-Day Grace Period:** Delete after 30 days of inactivity
- **Export Required:** User must download CSV before cancellation

Recommendation: Retain forever (increases re-subscription likelihood)

12.3 Social Features

Question: Should users be able to see other collectors' profiles?

Considerations:

- **Pro:** Community building, engagement
- **Con:** Privacy concerns, competition
- **Middle Ground:** Opt-in public profiles

Recommendation: Phase 4 feature with opt-in

12.4 Mobile App vs. Web App

Question: Should we build a native mobile app?

Analysis:

| Approach | Pros | Cons |

|-----|-----|-----|

| **Web-Only** | No app store approval, easier updates | Less engagement |

| **PWA** | Installable, offline support | Limited native features |

| **Native App** | Best UX, push notifications | High development cost |

Recommendation: Start with responsive web app (PWA in Phase 3)

12.5 Inventory Integration

Question: Should onboarding show real-time stock availability?

Scenario: User marks 100 elements as missing, but 20 are out of stock

Options:

- **Show All Elements:** User can wishlist out-of-stock items (frustrating if never available)
- **Hide Out-of-Stock:** Only show available elements (limits wishlist value)
- **Show with Badges:** Display "Out of Stock" label (transparent but cluttered)

Recommendation: Show all, but add "Notify Me" button for out-of-stock wishlist items

13. Appendix

A. Glossary

| Term | Definition |
|------------------------|---|
| Collector Cabinet | Web app for tracking element collections |
| Cube of the Month | Monthly element subscription service |
| Onboarding | First-time user setup flow |
| Collection Format | Size/type of element (10mm, 25.4mm, etc.) |
| Owned Element | Element user has physically received |
| Wishlist Item | Element user wants to receive as gift |
| Assignment Engine | Algorithm that selects next shipment |
| Subscription Detection | Process of linking Shopify subscription to user |
| Share Token | Unique URL identifier for wishlist sharing |

B. Referenced Documents

- `PROJECT_SUMMARY.md` – Overall project architecture
- `ASSIGNMENT_LOGIC.md` – Shipment selection algorithm details
- `SHOPIFY_INTEGRATION_GUIDE.md` – Technical integration steps
- `DEVELOPER_INTEGRATION_CHECKLIST.md` – Pre-launch validation

C. User Research Insights

Findings from 25 beta tester interviews:

1. **“I don’t trust automated systems with my collection”**
 - **Solution:** Show preview of first 3 shipments during onboarding
2. **“I collect multiple formats—this forces me to choose one”**
 - **Solution:** Phase 2 multi-format support
3. **“Marking 50+ elements manually is tedious”**
 - **Solution:** Order history import + bulk selection by group
4. **“I want my family to see what I already own, not just my wishlist”**
 - **Solution:** Wishlist view shows green (owned) + yellow (wished)
5. **“What if I mark an element by mistake?”**
 - **Solution:** Easy undo on dashboard (click owned element → “Remove from Collection”)

D. Design Mockup Links

(To be created by design team)

- Figma: Onboarding Flow Screens 1-5
 - Figma: Wishlist Public View
 - Figma: Dashboard First-Time Tooltips
 - Video: 30-second Demo for Welcome Screen
-

E. Technical Dependencies

Third-Party Services:

- **Shopify API** - Customer auth, subscription data
- **SendGrid/Klaviyo** - Email automation (welcome emails)
- **Sentry** - Error tracking
- **Google Analytics** - Funnel analysis
- **Mixpanel/Amplitude** - Event tracking (optional)

Internal Systems:

- **SKU Mapping Database** - Maps atomic numbers to product IDs
 - **Assignment Engine** - Separate service/module for shipment logic
 - **Admin Dashboard** - Staff interface for monitoring
-

14. Approval & Sign-Off

Document Prepared By: Abacus AI Agent

Reviewed By: (Pending)

Approved By: (Pending)

Approval Date: (Pending)

Next Steps:

1. Review and approve requirements
 2. Design mockups for Screens 1-5
 3. Set up Shopify subscription products
 4. Begin Phase 1 development
 5. Recruit 100 beta testers
-

End of Document